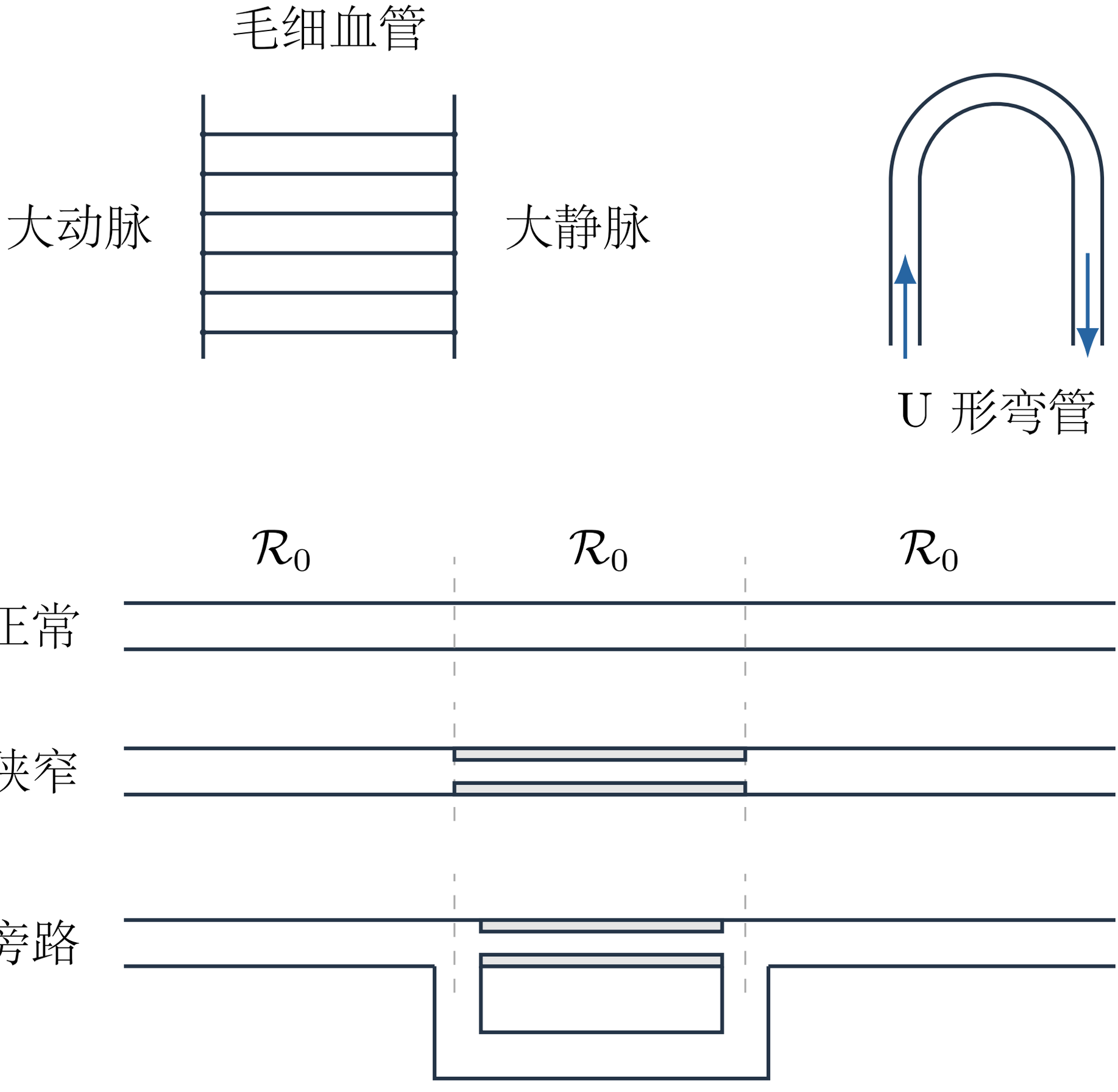
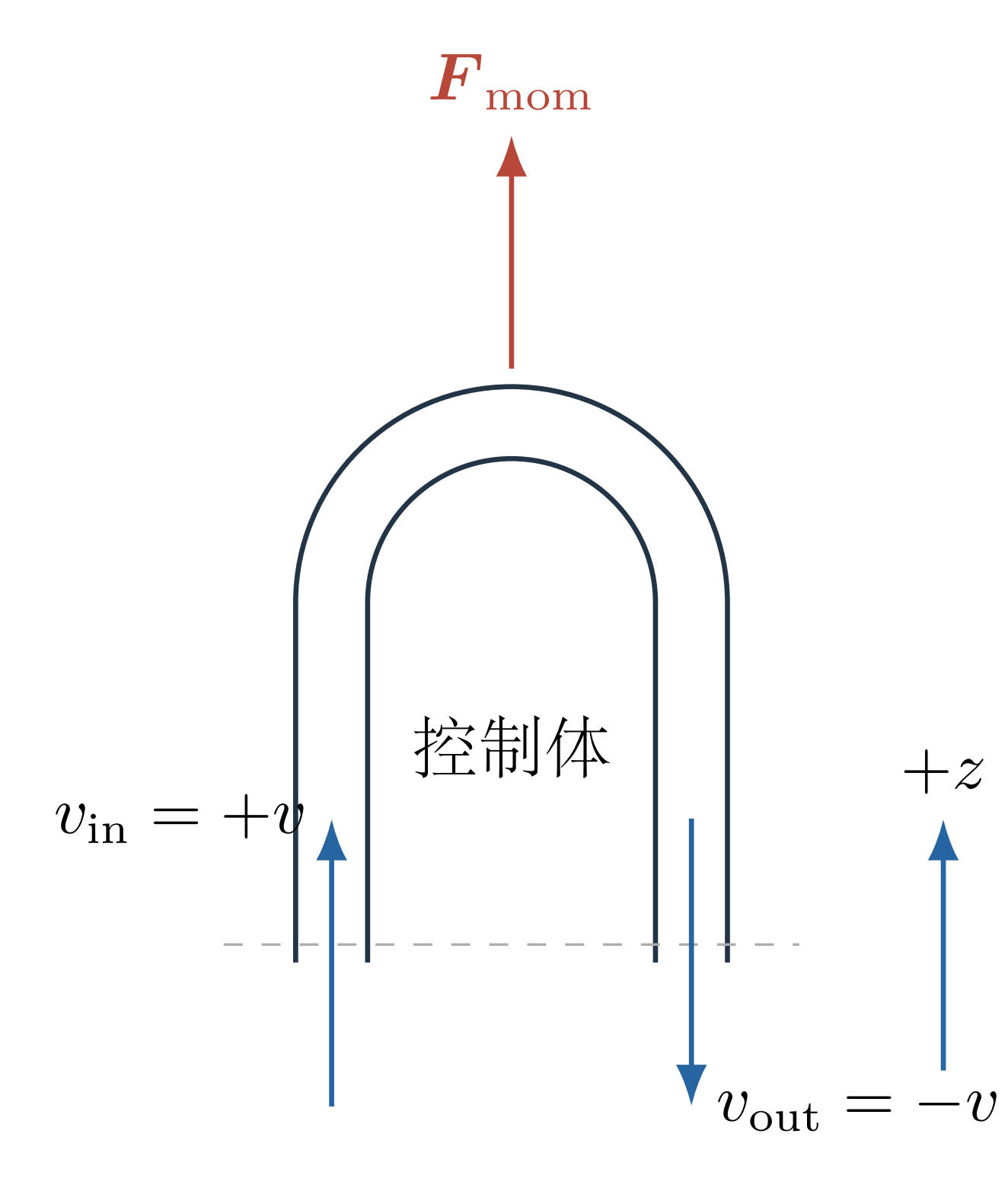
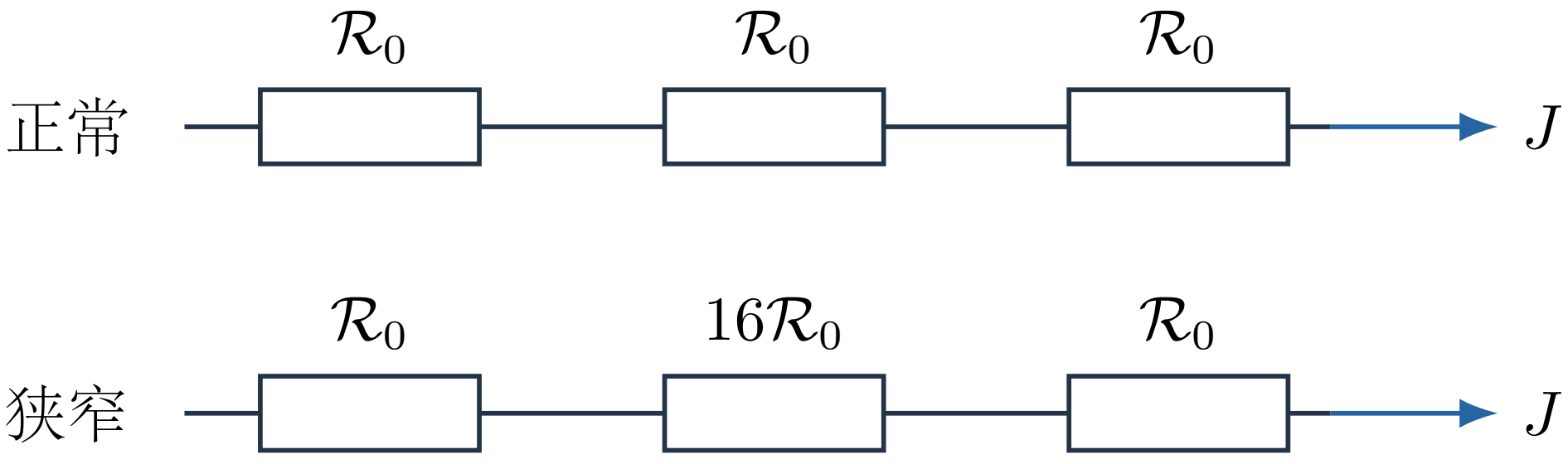
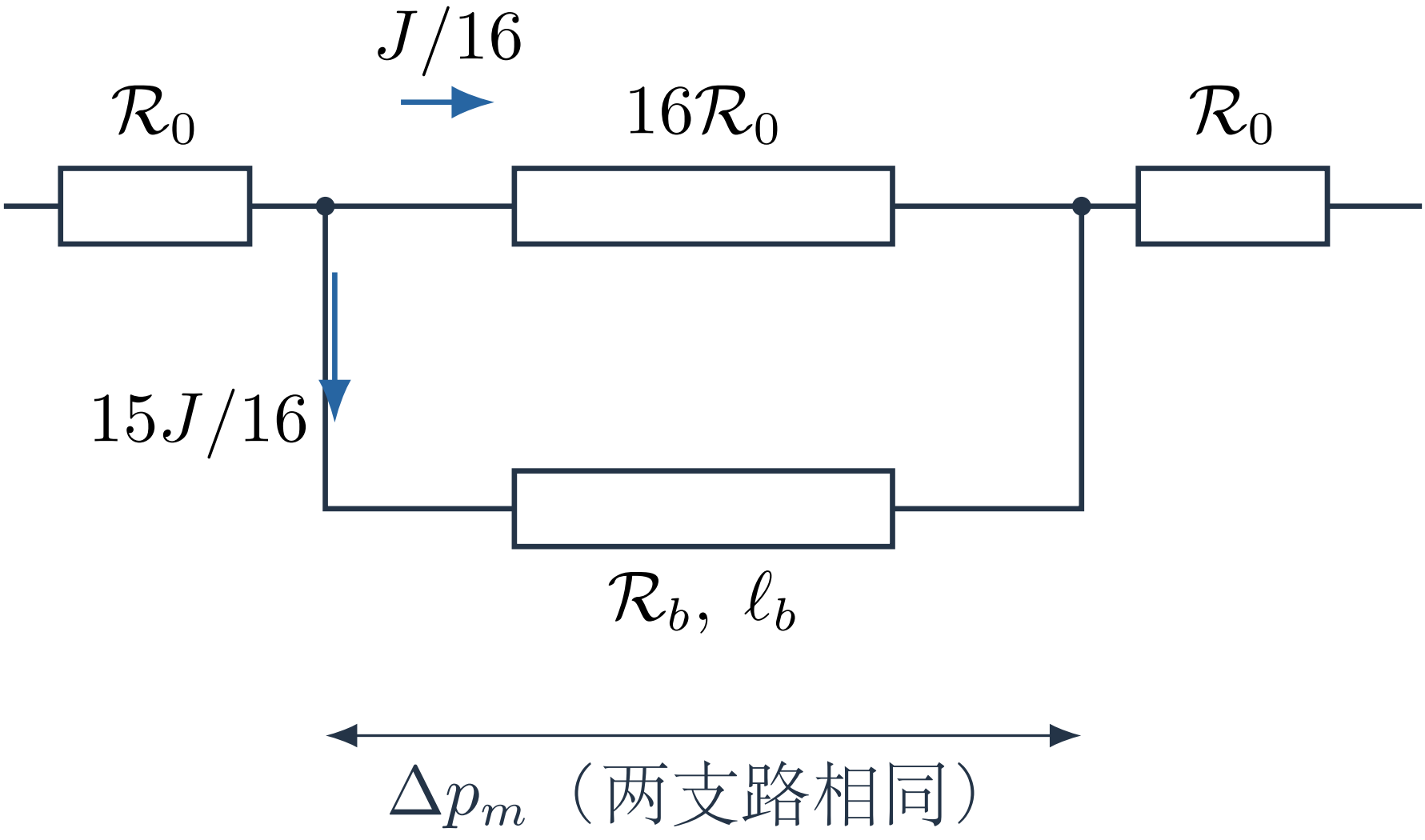
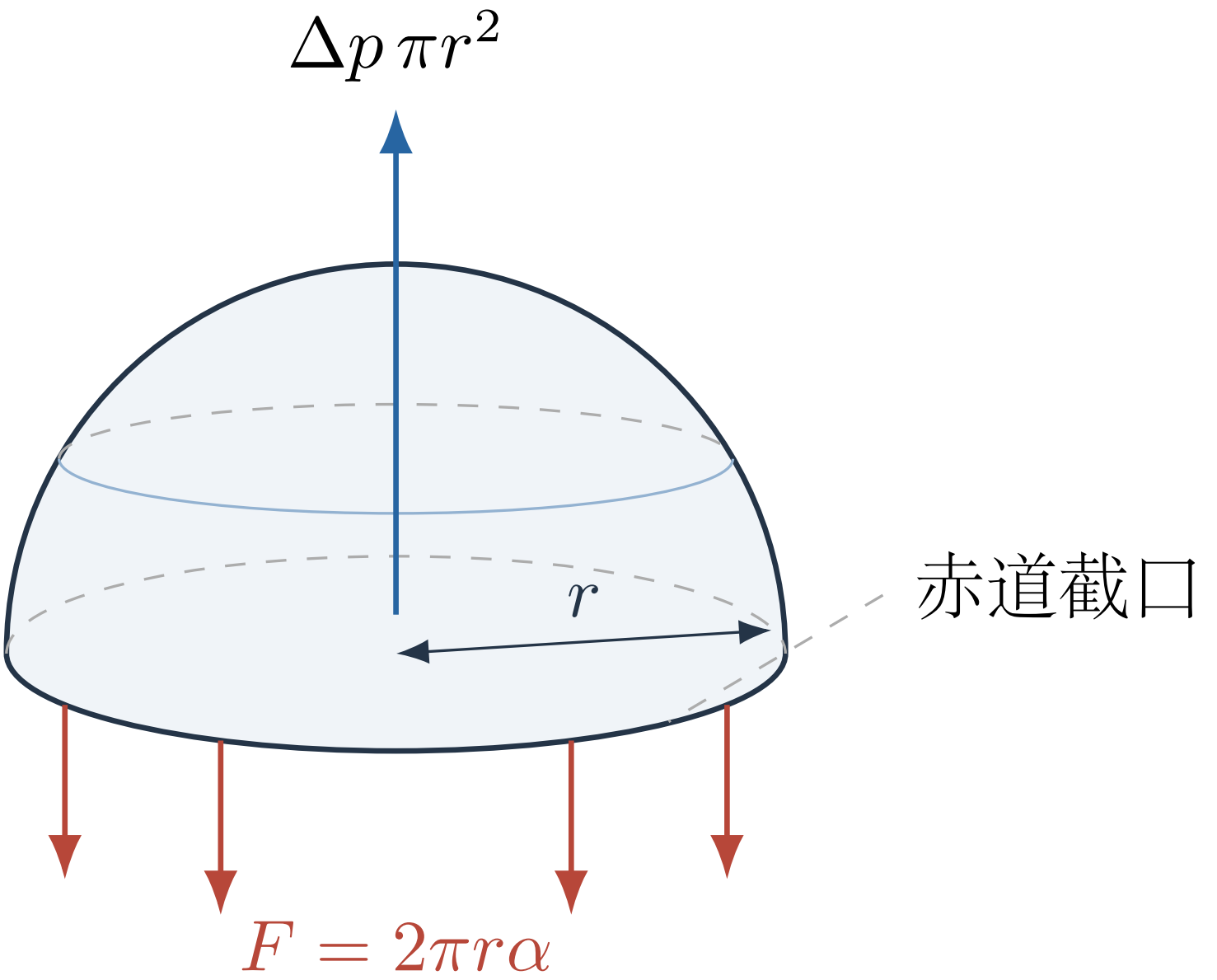

毛细血管
大动脉
大静脉
U 形弯管
$\mathcal{R}_0$
$\mathcal{R}_0$
$\mathcal{R}_0$
正常
狭窄
旁路


$\boldsymbol{F}_{\mathrm{mom}}$
控制体
$v_{\mathrm{in}} = +v$
$+z$
$v_{\mathrm{out}} = -v$


$\mathcal{R}_0$
$\mathcal{R}_0$
$\mathcal{R}_0$
正常
J
$\mathcal{R}_0$
$16\mathcal{R}_0$
$\mathcal{R}_0$
狭窄
J


$J/16$
$\mathcal{R}_0$
$16\mathcal{R}_0$
$\mathcal{R}_0$
$15J/16$
$\mathcal{R}_b,\ \ell_b$
Δp_m（两支路相同）


$\Delta p\,\pi r^2$
r
赤道截口
$F = 2\pi r\alpha$